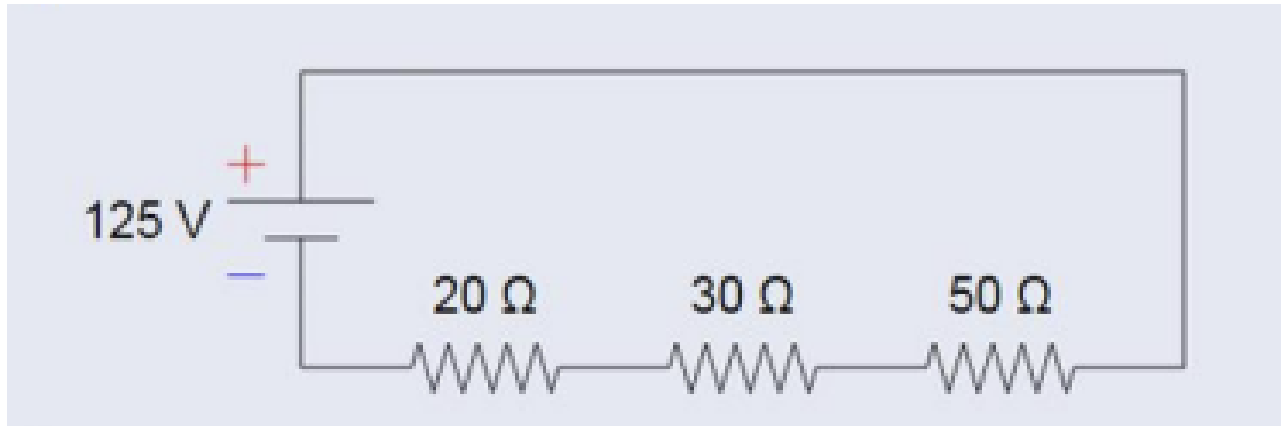




Givens:

$$V = 125V$$

$$R1 = 50\Omega$$

$$R2 = 30\Omega$$

$$R3 = 20\Omega$$

QN1. Total Resistance (Ohms)

$$R_{total} = R1 + R2 + R3$$

$$R_{total} = (50 + 30 + 20)ohms$$

$$R_{total} = 100ohms$$

QN2. Total Current (I)

$$I = \frac{V}{\Omega}$$

$$I = \frac{125V}{100\Omega}$$

$$I = 1.25A$$

QN3. Current Per Resistor

$$I_1 = I_2 = I_3 = 1.25A$$

QN4. Voltage Drop Per Resistor

$$V_1 = 1.25A * 20\Omega = 25V$$

$$V_2 = 1.25A * 30\Omega = 37.5V$$

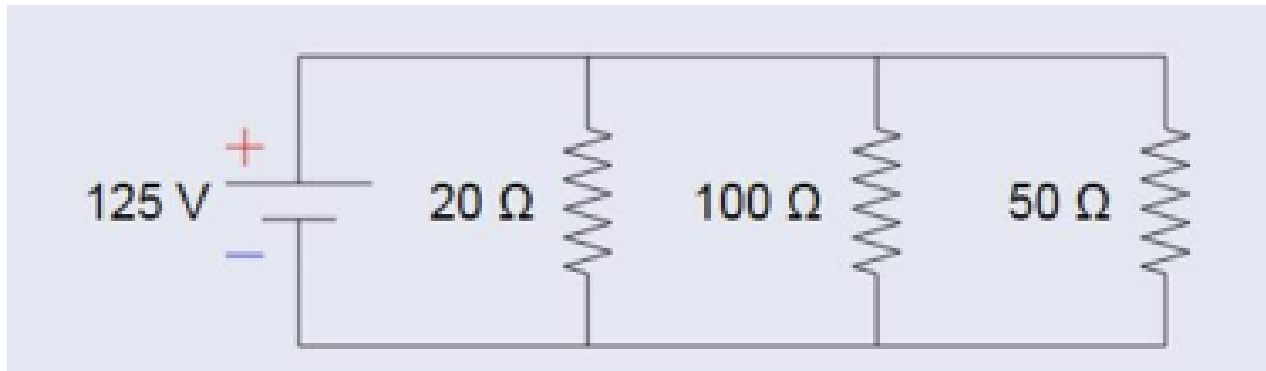
$$V_3 = 1.25A * 50\Omega = 62.5V$$

QN5. Power Dissipated Per Resistor

$$P_1 = 25V * 1.25A = 31.25W$$

$$P_2 = 37.5V * 1.25A = 46.875W$$

$$P_3 = 62.5V * 1.25A = 78.125W$$



Givens:

$$V = 125V$$

$$R1 = 20\Omega$$

$$R2 = 100\Omega$$

$$R3 = 50\Omega$$

QN1. Total Resistance

$$\frac{1}{R_{total}} = \frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3}$$

$$\frac{1}{R_{total}} = \frac{1}{20\Omega} + \frac{1}{100\Omega} + \frac{1}{50\Omega}$$

$$\frac{1}{R_{total}} = \frac{8}{100}$$

$$R_{total} = \frac{100}{8} = 12.5\Omega$$

QN2. Total Current

$$I = \frac{V}{\Omega}$$

$$I = \frac{125V}{12.5\Omega} = 10A$$

QN3. Current Through Each Resistor

$$I1 = \frac{125V}{20\Omega} = 6.25A$$

$$I2 = \frac{125V}{100\Omega} = 1.25A$$

$$I3 = \frac{125V}{50\Omega} = 2.5A$$

QN4. Voltage Drop Per Resistor

$$V1 = V2 = V3 = 125V$$

QN5. Power Dissipated Per Resistor

$$P1 = 125V * 6.25A = 781.25W$$

$$P2 = 125V * 1.25A = 156.25W$$

$$P1 = 125V * 2.5A = 312.5W$$